The cocktail study against the single-compound literature

How Argoul 2026 was run, which comparisons it supports, and why six were dropped

1. What the experiment was

Argoul and colleagues dosed **eleven PFAS as a single mixture** to 108 female C57BL/6 mice, 12 weeks old, 20.0 ± 0.8 g (Environmental Research 303:124802). Fifty-one animals received the mixture intravenously by lateral tail vein at 5 mL/kg; forty-nine received it by oral gavage at 10 mL/kg; the vehicle was saline with 10% ethanol for the intravenous route and 5% for the oral. Doses were set per compound from a preliminary study so that each would be measurable while staying in a low molar range - 0.019 to 1.55 mg/kg, tracking persistence rather than sweeping a dose-response.

Sampling was destructive and ran to **119 days**: blood from three to five mice at each of fifteen times, from 15 minutes to 119 days. From 24 hours on, three mice per time per route went into metabolism cages for complete 24-hour urine and faeces collection, so the renal and faecal routes were measured rather than inferred. Eight untreated mice established background. Quantification was LC-MS/MS against seven isotope-labelled internal standards.

Plasma, urine and faeces were then fitted **simultaneously** by nonlinear mixed effects, Laplacian engine, with precision from bootstrap (n = 100 or 200). Clearance and steady-state volume are primary estimates; mean residence time and terminal half-life are derived from them. Two inputs come from outside the experiment and matter later: glomerular filtration rate is taken as 10 L/kg/day from the physiological literature, not fitted, and the unbound fractions are from a companion study in the same strain, not the same animals.

2. Why the comparison is worth making

This study carries unusual weight in the review: it is the only source that measures clearance and volume for many PFAS in one experiment, which is what the sex-versus-species decomposition and the structure-activity endpoint both need. Every other mouse value available comes from a study that dosed one compound. So the question is whether dosing eleven at once, at low dose, by mixed-effects fitting, reproduces the one-at-a-time literature.

3. Which comparisons were kept, and why six were not

Thirteen comparisons survive, against female-mouse studies that dosed the compound alone. Six were dropped, all for the same reason: **no single dose can be attributed to the reported parameter**, so a dose column alongside them could not be read as a dose.

dropped	rows	why
Lou 2009 (PFOA)	3	one 2-compartment fit pooled across 1 and 10 mg/kg
Tatum-Gibbs 2011 (PFNA)	2	a half-life *range*, 25.8-68.4 d, spanning both doses rather than a value per arm
US EPA PFHxA IRIS	1	a $V_{d,\beta}$ computed from Gannon 2011, whose mouse arm ran 2 *and* 100 mg/kg

A seventh exclusion is of a different kind. The EPA pooled Bayesian fit was also kept out, because it re-analyses the same single-compound studies: placing it beside them would count the same animals twice, and it is a model output rather than a measurement - for PFBA it returns 0.186 d where every Chang arm sits at 0.116-0.128.

The exclusions cost coverage, which is worth stating plainly. PFOA loses its second comparator; **PFNA and PFHxA leave the comparison entirely**. PFHxA was the one outlier in the other direction (5.13× high) and the only compound the study places in net tubular secretion, so its kinetics differ in kind regardless. They cost almost nothing in the answer: including the three dropped half-life rows gives ten comparisons at 0.99× against seven at 1.04×.

4. The result

parameter	n	compounds	geometric mean	span
clearance	3	PFOA, PFHxS	**0.40×** (2.5× low)	0.34-0.49×
volume of distribution	3	PFOA, PFHxS	**0.54×** (1.8× low)	0.42-0.65×
half-life (their ratio)	7	PFBA, PFHxS, PFOS	**1.04×**	0.79-1.30×

Read from db/argoul_vs_single_compound.csv at build time; references in the _sources sidecar. Half-life is ln2·MRT, since the study reports mean residence time rather than a terminal half-life.

Both absolute terms sit low by a similar factor while their ratio does not. That is the signature of a **shared multiplicative scaling**, not a disagreement about elimination: in $\ln 2 \cdot V_d/CL$ a factor common to both cancels. One caveat on how much of a check that is - volume is usually not measured apart from clearance, since non-compartmental analysis gives $V_z = CL/\lambda_z$, so for the Sundström rows 'both low' is largely one observation seen twice. Fujii is the exception, with volume from dose/C(0) and clearance from dose/AUC, and there the two are low independently (0.38× and 0.59×).

5. What explains the offset

Dose is the obvious candidate and it does not survive testing. Matched per compound, the dose gap fails to order the discrepancies: Fujii dosed PFOA at 0.128 mg/kg against the cocktail's 0.080, a gap of 1.6×, and disagrees *most* (0.38×), while Lou at a gap of at least 12× agrees *best* (0.75×). The log-log slope once quoted in support was fitted through the cocktail's own point, which carries 79% leverage on it; dropping it and extrapolating Sundström's two doses alone predicts 2.04 mL/kg/d against 1.30 observed, leaving about 1.6× unexplained. Dose contributes; it is not the explanation.

Strain is a confounder the review had not isolated. Argoul used C57BL/6; every comparator is CD-1 except Fujii, which is FVB/NJcl. All thirteen comparisons therefore cross strains, and no study in hand separates strain from cocktail, dose or method. The comparison CSV now carries both strains per row so the confound is visible rather than implicit.

Cocktail competition for shared reabsorptive transporters would raise clearance, not lower it, so it argues against the observed direction unless the competition is for plasma binding sites instead. **Mixed-effects shrinkage** compresses spread rather than shifting a centre, so it is the weakest of the four.

6. What this does and does not cost the review

The within-experiment contrasts are **unaffected**: a shared scaling cancels from a ratio, so the 5,254× clearance span against 7.7× in volume across the eleven compounds stands whatever the common factor turns out to be. The absolute clearances are what the renal handling ratio R is built from, and those carry the roughly 2.5× low bias - a flat –0.40 log unit offset on every compound. Being flat, it cancels from every between-compound contrast the endpoint actually claims, and the nearest reabsorbed compound sits 0.98 log units from the R = 1 divide, so the bias would have to be 9.6× rather than 2.5× to reclassify anything. What it does cost is the cross-species comparison, where an absolute mouse value meets an absolute human one: read those gaps as carrying a factor of about 2.5 on top of the unbound-fraction uncertainty, understating the mouse.

The experiment that would settle it is not another comparator. It is the same cocktail at a second total dose, in a strain the single-compound literature also used - which would separate dose, cocktail and strain in one design.

Reproduce: scripts/argoul_vs_single_compound.py (comparison), scripts/argoul_dose_matching.py (the dose test), scripts/make_argoul_memo.py (this memo). Full treatment in report/REPORT.md section 11.3a and section 10 of the main review.